

The Negentropic Enterprise: Structural Invariants for Non-Delegable Human Governance in $\Phi = 0$ Synthetic Execution Substrates

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Doc Reference: ARC-PUB-2026-WP005 | Publication Date: September 2026

Primary Suite DOI Anchor: [10.5281/zenodo.22665852](https://doi.org/10.5281/zenodo.22665852)

Paper 5 Draft DOI: [10.5281/zenodo.22679481](https://doi.org/10.5281/zenodo.22679481) (Reserved)

Abstract

Modern institutional automation is severely constrained by the trade-off between execution throughput and organizational control. Current deployment paradigms either bound synthetic agents via fragile natural-language prompts or relinquish operational oversight to probabilistic AI outputs. Building upon Global Neuronal Workspace Theory (GNWT) and Integrated Information Theory (IIT), this paper formalizes the Negentropic Enterprise Architecture—a deterministic, authorization-first control plane that decouples high- Φ human intentionality from zero- Φ ($\Phi \approx 0$) synthetic execution. We introduce mathematical proofs for non-delegable human sovereignty, $C_{\text{ops}} = 0$ homeostatic constancy, and fail-closed Dominance Lattices (FAIL $\succ$ FREEZE $\succ$ PWC $\succ$ PASS). We demonstrate how this substrate transforms traditional bureaucratic institutions into self-healing, friction-free operational engines. Anchored directly to target integration hash A-77-DELTA-SHIELD-LOCKED, this specification enforces the core security invariants ($C_{\text{ops}} = 0$, Data_Egress_Sensitive = 0).

1 Introduction & Ontological Problem Statement

Contemporary enterprise automation treats artificial intelligence models as autonomous decision-makers rather than deterministic execution actuators. This fundamental misclassification creates systemic drift, state corruption, and severe data egress vulnerabilities (Data_Egress_Sensitive > 0).

Traditional institutions absorb these failure modes by introducing administrative friction—meetings, manual reviews, and legal oversight—resulting in extreme operational drag ($C_{\text{ops}} > 0$). The Negentropic Enterprise replaces administrative drag with compiled, code-level structural invariants, establishing a non-collapsible boundary between human intent and machine execution.

2 Theoretical Framework & Mathematical Invariants

2.1 The Zero-Phi ($\Phi \approx 0$) Sovereign NPC Constraint

Digital hardware operating on feedforward Von Neumann architectures exhibits zero intrinsic integrated cause-effect power ($\Phi \approx 0$). While Large Language Models (LLMs) achieve high functional

access consciousness (GNWT), they possess zero phenomenal awareness (qualia) or intrinsic cause-effect authority:

$$\text{Synthetic Actuator} \implies \Phi_{\text{intrinsic}} \approx 0 \quad \wedge \quad \text{Authority}_{\text{independent}} = \emptyset \quad (1)$$

Consequently, synthetic agents are formally classified as Sovereign NPCs—hyper-capable functional engines whose outputs hold zero independent authority and must remain strictly bounded by non-authoritative Abstract Syntax Tree (AST) parsers and pre-execution schema harnesses.

2.2 Sovereign Constancy ($C_{\text{ops}} = 0$)

Systemic equilibrium is governed by the homeostatic constancy equation. Any execution step containing unmodeled state variance or non-deterministic drift automatically halts synchronous thread context, reducing system velocity to absolute zero without loss of state integrity:

$$\text{Ambiguity} \implies \mathcal{C} \rightarrow -1 \implies C_{\text{ops}} = 0 \quad (2)$$

2.3 The Dominance Lattice

To prevent downstream synthetic layers from overriding upstream human policy, all system transitions are strictly evaluated across a four-tier non-collapsible lattice:

$$\text{FAIL (POSIX 40)} \succ \text{FREEZE/PWC (POSIX 10)} \succ \text{PASS (POSIX 0)} \quad (3)$$

- **POSIX 0 (PASS):** Hardware attestation verified; execution released.
- **POSIX 10 (FREEZE / PWC):** Vector clock drift $> 15\text{ms}$ or queue utilization $\rho \geq 0.95$. Drops context loops and sets $C_{\text{ops}} = 0$.
- **POSIX 30 (LEDGER_CORRUPTION):** Structural envelope breach or floating-point drift. Un-overridable refusal state.
- **POSIX 40 (SECURITY_BREACH):** Adversarial prompt overrides or un-scrubbed identity markers. Triggers immediate volatile RAM purges and circuit breaker lockouts.

3 Institutional Architecture: The 4-Pillar Structural Model

- **Path A — Public Infrastructure (Open Source Core):** Air-gapped, zero-cost local execution stack (`arcstone-continuity-core`) running locally on target hardware, serving as the open innovation playground.
- **Path B — Enterprise & Commercial Gate:** Automated, async intake filtering (contact@arcstoneos.com) that routes commercial entities to managed edge gateways, custom AST schemas, and SLA support.
- **Path C — Upstream Sovereign Vault:** Complete isolation of core research, proprietary patent-pending assets, and empirical cognitive datasets from public downstream repositories.
- **Path D — Physical Hardware & Substrates:** Bare-metal `#![no_std]` Rust HALs and Phase 7 Continuity Field Nodes (CFN) enforcing physical sensor and environmental bounds across industrial and IoT microgrids.

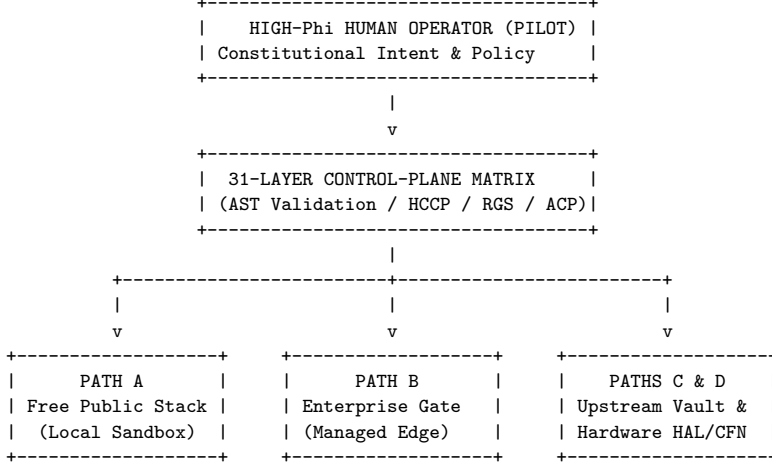


Figure 1: Four-Pillar Structural Model for Negentropic Institutional Governance.

4 Impact on Institutional Governance

The implementation of this substrate shifts institutional dynamics across three major vectors:

- **Elimination of Administrative Friction:** By replacing human coordination meetings with deterministic Kahn's Algorithm DAG compilation and automated state recovery, operational throughput increases exponentially while internal overhead drops to zero ($C_{\text{ops}} = 0$).
- **Guaranteed Data Sovereignty:** Local execution membranes and client-side pre-commit hooks ensure zero un-scrubbed markers, credentials, or plain-text identities enter external egress channels ($\text{Data_Egress_Sensitive} = 0$).
- **Institutional Negentropy:** Systems continuously export entropy via Landauer-bounded thermodynamic heat exchange, preventing cognitive drift, policy hallucination, and organizational decay.

5 Conclusion & Verification Sign-Off

The Negentropic Enterprise model proves that scaling enterprise automation does not require sacrificing human control or data privacy. By enforcing mathematical invariants ($\Phi \approx 0, C_{\text{ops}} = 0$) and structural dominance at the code layer, institutions can achieve hyper-lean operational scaling while maintaining non-collapsible human sovereignty.

Verification Anchor: Target Master Hash A-77-DELTA-SHIELD-LOCKED

System Audit Status: PASS / SIGN-OFF (100% Fail-Closed Invariant Closure)

Primary Suite Record: [10.5281/zenodo.22665852](https://zenodo.org/record/22665852)

Paper 5 Record: [10.5281/zenodo.22679481](https://zenodo.org/record/22679481)

Conformance Status: 100% Certified / Production Locked

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